

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Driven by a problem solving mindset and fueled by curiosity, I always aspire to develop scalable solutions for the real-world, impactful problems.

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary

Supervisor: [Márton Vaitkus](#)

Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia

GPA: 3.2/4.0 (**rank 3/34**)

Supervisor: [Galtbayar Artbazar](#)

Thesis: [Evaluating land degradation using image processing methods](#)

Presentation

EXPERIENCE

Center of Mathematics Application Research Lab Oct, 2023 – June, 2025

Undergraduate Lab Lead & Research Intern

- **Lab Lead:** Selected as Undergraduate Lab Lead amongst the 6-8 student team in the final year. Co-taught and discussed a machine learning topics in weekly seminar meetings.
- **Research:** Developed a custom, vendor specific **Python program** which takes camera and sensor parameters into account. This enabled us to reach a conclusion that the spatial clustering of weed can be captured using Moran's I coefficient performed on images.
- **Organization:** Implemented a unified, documented folder structure for the research data backup which enabled efficient data retrieval and writing procedure.

PROJECTS

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Self-initiated project to reproduce key examples from C. Bishop and H. Bishop's *Deep Learning Foundations and Concepts* to gain theoretical and fundamental understanding of **polynomial regression** and **probability theory**.

[aiSim Data Integration for 3D Gaussian Splatting](#)

- Implemented a custom JSON parser to extract and sensor data obtained by aiSim, an autonomous driving simulator, into a nerfstudio format.

SKILLS

Programming: Python, Julia, C/C++, R, MATLAB

Tools: Docker, Linux, Bash, Git, $\text{\LaTeX}$